01 A distinct identity

A practical UI and UX specification for your FYP product. Prepared 26 October 2026.

CastraVolt

EV INFRASTRUCTURE PLANNING

Midnight Iris



Recommended direction

Deep ink, electric violet, soft lilac and a restrained rose accent give CastraVolt an identity beyond the familiar green EV aesthetic. Light analytical surfaces keep maps, numbers and long sessions comfortable to read.

This guide specifies colour codes, typography, exact placement, screen content, map behaviour, component states, responsive layouts and a realistic build sequence. It is a design recommendation, not a change to your research methodology.

Study boundary: Kuala Lumpur and Selangor. Design concepts are illustrative. Only use real study results in the implemented dashboard; confirm the final prediction target with your supervisor.

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02 Colour system

Theme name: Midnight Iris. Use one coherent palette across navigation, maps, charts and reports.

Midnight Ink
#19172B
Sidebar, headings and logo backgrounds.

Electric Iris
#6546D7
Primary actions, selected sites and brand focus.

Signal Rose
#B83D70
Small comparison accents; never a second primary colour.

Token / hex	Role and placement
Canvas #F7F6FB	Overall page background; separates white cards.
Surface #FFFFFF	Cards, tables, filters and detail panels.
Iris wash #EEE9FC	Selected rows, light badges and contextual notes.
Iris hover #5033B4	Primary button hover and pressed emphasis.
Heading / text #282538	Main body copy, numbers and form labels.
Secondary text #625D73	Subtitles, helper text and axis labels.
Divider #DDD9E8	Subtle separators; not sufficient alone for an input outline.
Control border #8A829D	Input edges and important interactive boundaries.
Supporting blue #285BC4	Observed series or an alternate analytical category.
Error #AD2947	Validation errors with explanatory text.
Success #42614D	Completed actions only; a muted functional accent, not the brand.

Balance: about 75% neutral surfaces, 20% ink and iris, and up to 5% rose or supporting colours. Avoid full-page gradients, neon glows and many competing coloured cards.

03 Typography and geometry

Use a disciplined layout so the dashboard looks like one product, rather than separate charts.

Typography

Element	Specification	Use
Page title	28 px / 36 px, 600	One main title per screen.
Section title	20 px / 28 px, 600	Map, charts and panel sections.
Body / form label	14 px / 20 px, 400 / 500	Main interface text.
Table content	14 px / 20 px, 400	Aim for readable analytical detail.
Supporting text	12 px / 18 px, 400	Dates, units and short notes.
KPI value	32 px / 40 px, 600	Tabular figures for stable alignment.

Recommended family: Inter for headings and body; system-ui, Segoe UI and sans-serif as fallbacks. Use weights 400, 500 and 600. Reserve monospace for technical identifiers on the data page.

Desktop dimensions - design frame 1440 x 900 px

Region	Placement and measurements
Sidebar	224 px wide, fixed left; 24 px inner padding.
Top utility bar	64 px high; align content to the main page grid.
Main content	32 px outer padding; maximum width 1600 px on wide screens.
Grid	12 columns; 24 px gutters. Map row: 8 columns + 4 columns.
Spacing scale	4, 8, 12, 16, 24, 32, 48 px. Use 24 px between major cards.
Shapes	12 px card radius; 8 px inputs/buttons; 6 px small badges.
Elevation	Default 1 px border; floating panels: 0 8px 24px rgba(25,23,43,.10).

Keep content left-aligned. Right-align numerical columns. Cards use 20-24 px padding. A page may scroll vertically; do not shrink charts or text to force everything into one viewport.

04 Navigation and user flow

Primary user: a planner, researcher or supervisor exploring evidence about charging infrastructure.

Navigation	Question it answers	Core content
Overview	What is in the study area?	KPIs, map, distribution charts.
Site explorer	What surrounds a particular site?	Search, map, site detail and POIs.
Model analysis	How well does the model perform?	Target, metrics, error analysis and model comparison.
Data & methods	Where did these results come from?	Source coverage, cleaning notes, target and limitations.

A simple end-to-end journey

1. Open Overview

Start with the full KL + Selangor study area. Read the source date and coverage before interpreting the totals.

2. Narrow the area

Choose state, then local authority and site category. The map, KPIs and charts follow the same active filters.

3. Inspect a site

Click a map marker or table row. The matching site becomes selected and the detail panel opens.

4. Review evidence

Inspect infrastructure and POIs. Open Model analysis to see what the model predicts and its evaluation.

5. Export or reset

Download the filtered table, with units and source period, or reset to the study-wide view.

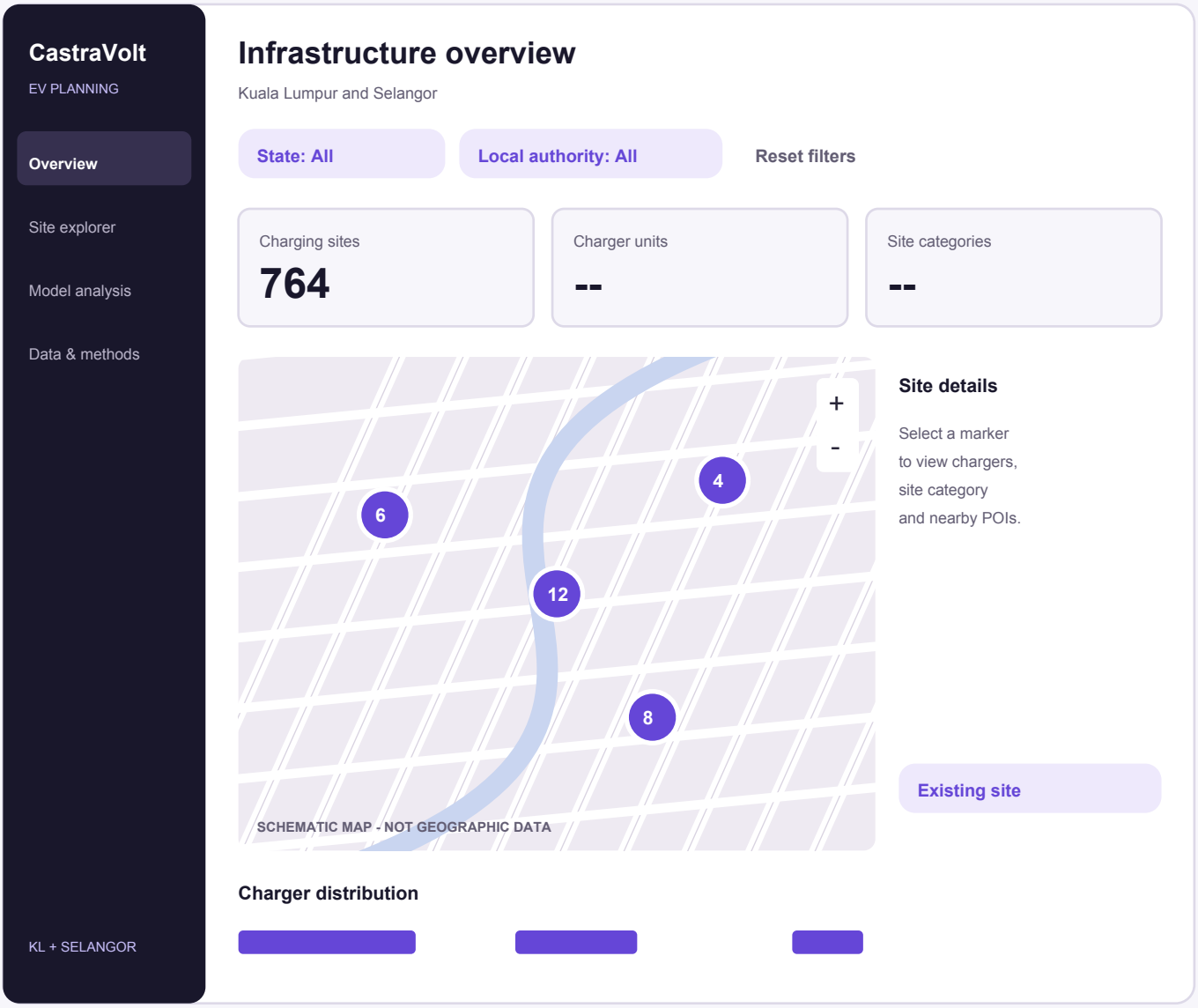
Sidebar and header

Logo at top left; descriptor underneath. Navigation starts 32 px below it. Active item: #302B47 background, white text and a 3 px lilac left indicator. Keep the study-area label at the bottom. Header right: Data & methods link and a compact source-period label.

> For the FYP prototype, open directly into the useful dashboard. Add login, billing or account management only if your actual project requirements need them.

05 Overview concept

A visual layout reference. This is a schematic interface, not a live dashboard or a geographic map.



Reading order

Title and study area > filters > summary cards > map and site detail > supporting charts. Keep the map visually dominant.

Visual emphasis

The dark sidebar anchors the product. Iris marks active controls and locations. Neutral cards allow numerical results to stand out without decorative colour blocks.

Concept values: 764 reflects the previously inspected charger file; other values are deliberately unfilled. Recalculate every KPI from the current cleaned dataset during implementation.

06 Overview placement rules

Build this screen first; it can demonstrate useful analysis before model results are available.

Area	Exact UI specification
Heading row	Title left: Infrastructure overview. Subtitle: Explore charging infrastructure across Kuala Lumpur and Selangor. Source date at right.
Filters	State, local authority, site category; optional AC/DC filter if its meaning is explicit. 44 px high; 16 px gaps. State selection updates available local authorities.
KPI row	4 equal cards: charging sites, recorded charger units, local authorities covered, median chargers per site. Show units and filter scope.
Map row	8 columns map + 4 columns detail panel; 480-560 px desktop map height. Map never hidden behind a fixed header.
Charts row	6 columns each: charger units by local authority and site count by category. Use horizontal bars, descending order, starting at zero.
Data context	Place source/snapshot information directly beneath each relevant figure. A file update date and a record reporting date are different concepts.

Metric definitions

Charging sites: count validated site records, with unique-site rules documented. Charger units: sum the verified existing-count field. Local authorities: distinct valid authorities in the filtered view. Median chargers per site: median existing count for the filtered records. Confirm whether the source counts ports, units or stations before naming the unit.

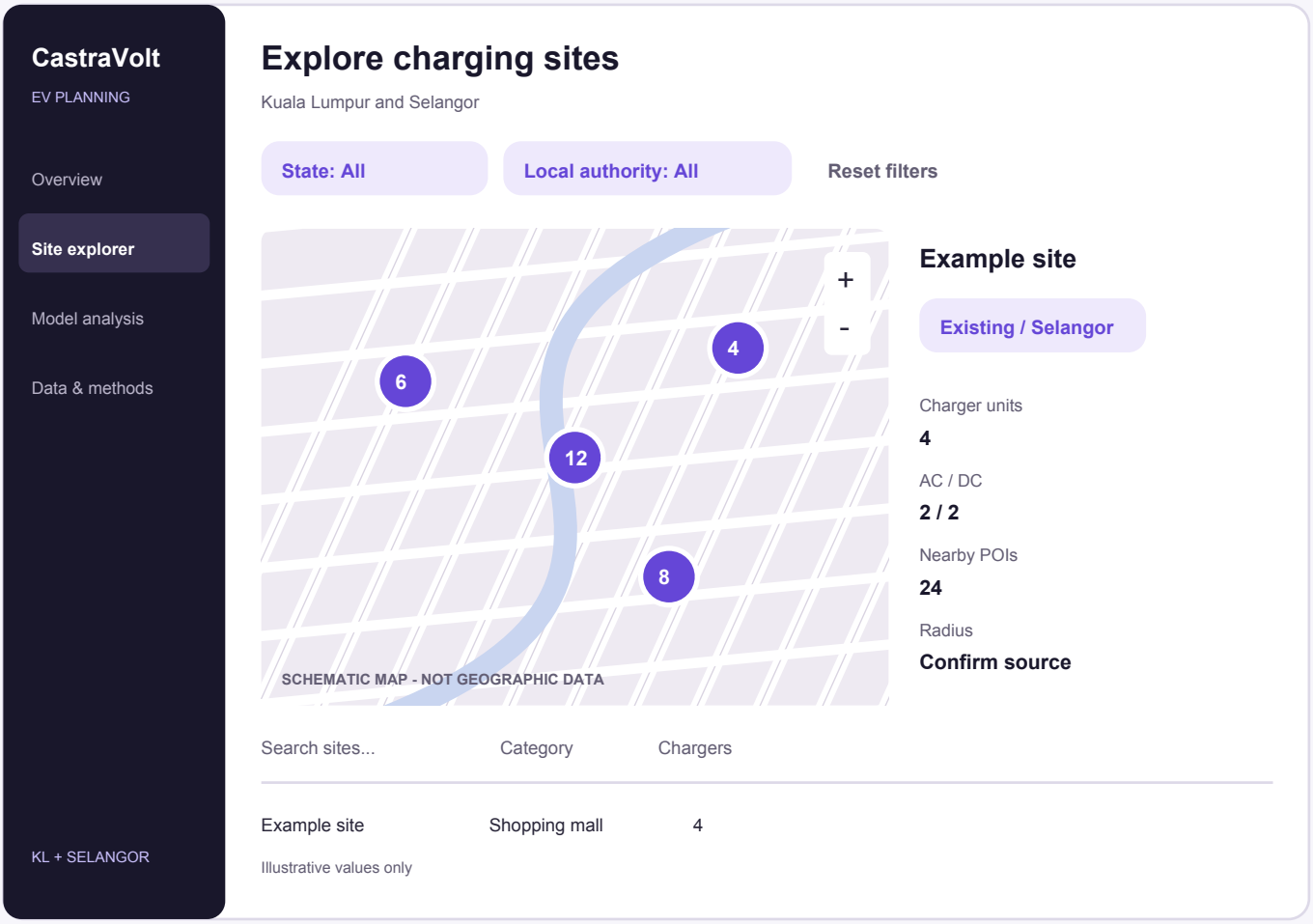
Filtering behaviour

Use an Apply filters button for a predictable update if processing is expensive. Show the selected values immediately, then update all visuals together. If a chart click filters the page, display a removable chip. Reset filters restores all study-area sites and clears the selected site.

> Do not add a national state selector. Registration trends can be a separate context view; they should not silently change site metrics or be treated as site-level demand.

07 Site explorer concept

The detail panel turns a map into a useful research tool. Keep one site selected at a time.



Detail order	Fields and display
1. Identity	Site name, category, local authority, state; status as a labelled badge.
2. Infrastructure	Existing count prominently; AC/DC and indoor/outdoor breakdown underneath. Proposed count separately, with its verified meaning.
3. Nearby facilities	Eight POI category counts in a 2-column grid. Display radius, extraction date and source when verified.
4. Provenance	Coordinates, record date and source. Use Not recorded for a missing value; never replace it with a zero.

Search by site name or local authority. Table columns: site, local authority, category, existing units and view details. Use 48-56 px rows, sortable headers and 25 rows per page. A selected row and its map marker must match.

08 OpenStreetMap experience

Use OpenStreetMap data with a deliberate display layer. OSM data access and map rendering are different parts.

Recommended first implementation

Use a Leaflet-based map for the prototype, with a suitable OSM-based tile source. Load your cleaned site records as an overlay. Use the official OSM API or Overpass only for appropriate data retrieval; they are not a replacement for a map renderer or a basemap tile service.

Basemap and styling

With standard raster tiles, roads, land and labels are already baked into the image. Style CastraVolt markers, boundaries and panels independently. If you need full control over road and land colours, choose an OSM-derived vector style with a compatible renderer and tile provider. Do not promise custom raster styling through the OSM API.

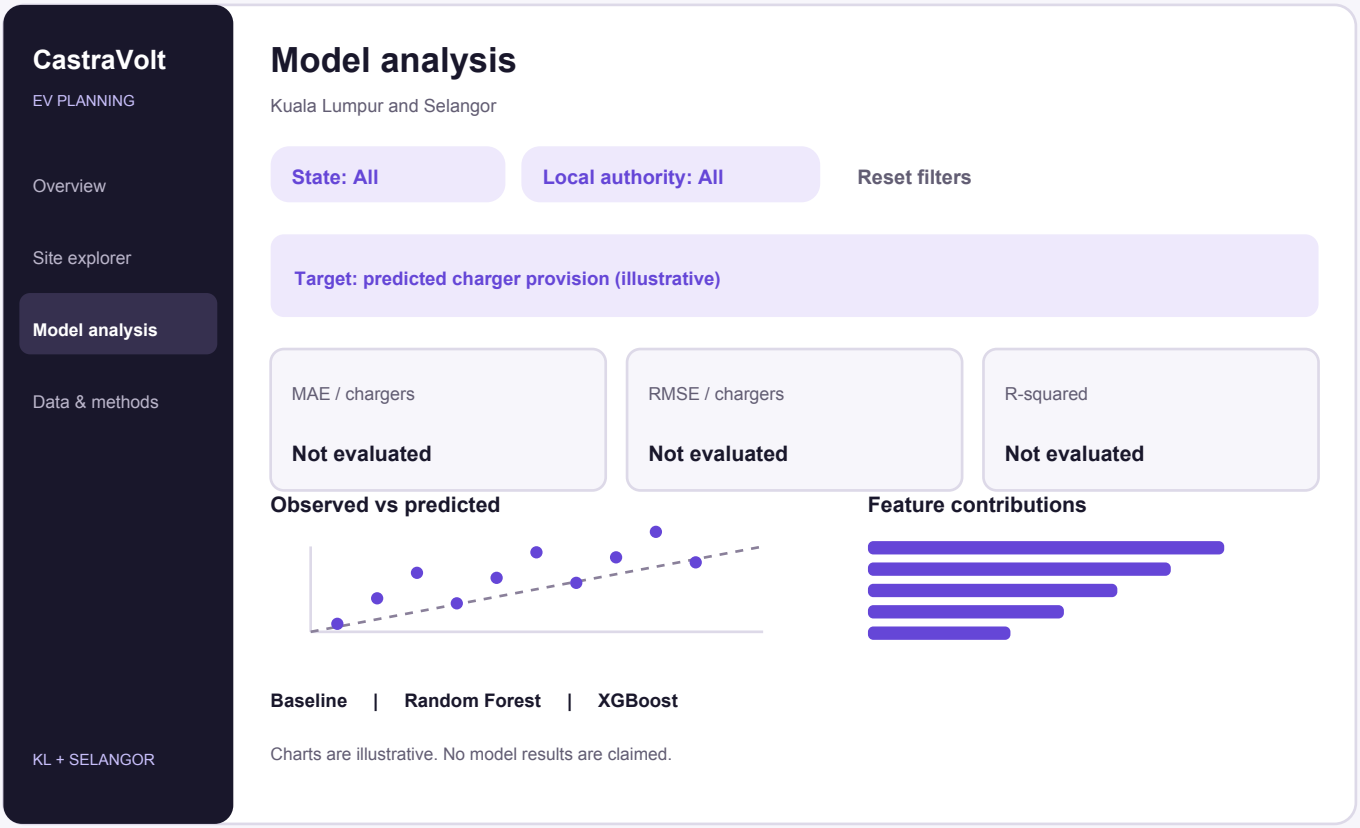
Layer / control	Recommended appearance
Existing site	Iris circle, #6546D7, 12-16 px; 2 px white border and dark outer edge.
Selected site	20 px iris marker; white border plus outer ink ring. Also show site name in the detail panel.
Cluster	32-44 px iris circle with white count. Size represents grouped site count, not demand.
Study boundary	#5033B4 line, 2 px; #6546D7 fill at 5% opacity. Use a verified boundary source.
POI radius	#285BC4 dashed outline, 2 px; faint fill. Label radius and distance method.
Controls	Zoom top left; layers and Fit study area top right; legend bottom left; attribution bottom right.

Start by fitting the verified KL/Selangor boundary or site bounds. Cluster dense markers. Selecting a cluster zooms in; selecting a site opens the panel. Keep the legend away from controls and do not trap page scrolling when the map gains focus.

Keep linked attribution visible: © OpenStreetMap contributors. Retain provider attribution too. Public OSM tiles require compliance with their usage policy; do not bulk-download or prefetch them for offline use. See official sources on page 14.

09 Model analysis concept

The research result must be understandable, with its target and evaluation visible before its visual impact.



Section	Required content
Context strip	Target name, unit, study population, feature set and evaluation split.
Metric cards	MAE, RMSE and R-squared. Include baseline comparison; no invented improvement percentage.
Comparison table	Baseline, Random Forest and XGBoost; same test sites, units and split. Identify chosen model and selection reason.
Charts	Observed versus predicted with equality line; error by area; feature importance labelled as association, not causation.

If your target is installed count, use Predicted charger provision. If your target is usage, name the measured unit, such as sessions/site/day. Predictions versus installed supply must not automatically become a shortage map. New-area recommendations need candidate-area data and independent validation.

Display Not evaluated until a model has actually been tested. The concept chart above is illustrative and contains no claimed model performance.

10 Registration context view

Place this as a tab within Data & methods, or a secondary section on Overview. It is adoption context.

Layout and content

Placement	Specification
Heading	EV registration trends. Subtitle states the exact vehicle scope, fuel filter and coverage.
Filters	Year, comparable period and registration-office area. Label the area field clearly.
Main chart	Monthly counts as lines; 2024 #285BC4, 2025 #6546D7, 2026 #B83D70. Add point symbols and direct labels.
Comparison cards	Same-period counts and changes only. Show dates explicitly, such as January-May, rather than implying a full year.
Coverage panel	First/last registration date, missing periods, source and inclusion rules for each file.
Exclusion note	Show excluded Rakan Niaga and other-area record counts, where relevant; link to the cleaning log.

Data interpretation in the interface

Registration geography

State labels identify registration-office locations or the partner portal, not verified owner residence. Use Registered at KL/Selangor offices rather than Resident EV population.

Rakan Niaga

Do not allocate portal registrations to KL or Selangor. Keep an explicit coverage note so the subset is not mistaken for the complete local fleet.

Historical coverage

The previously supplied 2026 registration file covered January-May. Inspect the actual vehicles_2026.csv file before setting the interface period; its coverage may differ.

Temporal consistency

Do not join later adoption values to earlier site records and label the result a genuine historical forecast. Keep sources and dates visible.

When comparing years, prefer point markers and line labels over colour alone. Missing months appear as gaps, not zero registrations. Tooltips show the exact period, count and scope.

11 Component specification

Reuse these measurements and states consistently on every screen.

Component	Default	Interaction / state
Primary button	44 px high; 16 px horizontal padding; #6546D7; white label; radius 8 px.	Hover #5033B4. Label names the action: Apply filters, Export CSV.
Secondary button	White fill; #6546D7 text; #8A829D outline; 44 px high.	Hover #EEE9FC. Avoid competing with the primary action.
Text input / select	44 px high; 14 px text; white fill; #8A829D border; visible label.	Focus: 3 px #6546D7 outline with 2 px white offset. Error: #AD2947 plus message.
Filter chip	32 px high; #EEE9FC fill; #5033B4 text.	Include clear/remove icon with accessible name and generous hit area.
Card	White fill; 12 px radius; #DDD9E8 border; 24 px padding.	No hover animation unless the whole card is interactive.
Table	48-56 px row; header #EEE9FC; 14 px text.	Hover #F7F6FB; selected #EEE9FC plus visible selection marker.
Badge	24-28 px high; short sentence-case label.	Existing: iris wash. Proposed: blue wash. Unknown: neutral. Status is not priority.
Toast	White surface; ink text; icon; close button.	Confirm completed action. Keep critical errors inline until resolved.
Tooltip	Ink surface; white text; max width 240 px.	Available on keyboard focus; never the only place critical information appears.

Analytical colour rules

Observed = #285BC4; predicted = #6546D7; residual emphasis = #B83D70. Use distinct line styles as well. If validated priority classes are introduced: low = #285BC4, medium = #6546D7, high = #B83D70; always show text labels. Do not reuse priority colours to imply charger operating status.

12 Accessible and responsive

Accessibility is part of the design specification. These checks are targets, not a claim of full conformance.

Readable combinations

Pair	Calculated contrast	Intended use
White / Electric Iris	6.16:1	Primary button labels.
Text / White	14.87:1	Main body and table text.
Secondary / Canvas	5.86:1	Helper text and captions.
Control border / White	3.65:1	Interactive input boundaries.

Aim for WCAG AA: 4.5:1 for normal text and 3:1 for large text; meaningful control boundaries and graphical objects generally need 3:1 against adjacent colours. Pale lilac is a background, not a small-text colour. Contrast ratios above cover these specific pairs only. [3, 4]

Responsive rules

Viewport	Layout behaviour
1280 px and above	224 px sidebar; 4 KPI cards; 8/4-column map and detail layout.
1024-1279 px	Sidebar can collapse to 72 px with labelled tooltips. KPIs use 2 x 2. Move details below map if cramped.
768-1023 px	Drawer navigation; 2-column cards; full-width map with detail panel below.
Below 768 px	Single column; 16 px padding; 360 px minimum map height; filters open as a full-width panel. Tables scroll inside their region.

Keyboard and chart access

Provide visible focus, logical tab order, labelled inputs and an Escape action for dialogs. Every map site must also be accessible from the table. Include a table alternative for charts. Use icons, line patterns and words with colour. Respect reduced-motion settings and test at 200% zoom.

13 Copy, feedback and identity

Short, specific language helps a supervisor or planner understand what the prototype actually shows.

Situation	Recommended interface copy
Initial site panel	Select a site on the map or in the table to view its details.
Loading data	Loading charging sites...
Applying filters	Updating results...
No matches	No sites match these filters. Clear filters to view the study area.
Missing source field	Not recorded
Map unavailable	The map could not load. Retry, or use the site table below.
Model unavailable	Model results are not available yet. You can still explore observed infrastructure.
Unknown POI radius	POI radius has not been verified. Interpret these counts with caution.
Export success	Filtered records exported.
Count-target label	Predicted charger provision
Evidence boundary	These predictions describe the modelled target and do not establish unmet charging demand.

Logo and brand expression

Use the CastraVolt wordmark in a medium or semibold geometric sans serif. Suggested symbol: an open C enclosing two offset vertical bars that suggest both charging ports and a location corridor. Create the mark as a simple vector; avoid fine detail, a generic green leaf or a crowded plug-and-car illustration.

Primary lockup: white wordmark on #19172B; lilac symbol. Light lockup: ink wordmark with iris symbol. Minimum symbol size: 24 px. Reserve clear space equal to half the symbol height. Keep the product name as CastraVolt, with capital C and V.

Motion and tone

Use 120-180 ms colour/opacity transitions. Keep map movement brief and optional. No flashing pins or continuously pulsing charger icons. Tone: calm, precise and supportive. Say what the data shows; avoid AI-powered accuracy claims without evidence.

14 Build plan and handoff

A practical delivery sequence, with a compact checklist for design review and implementation.

Phase	Deliverable and acceptance check
1. Foundation	Create colour/text tokens, sidebar, header, buttons and filter fields. Check contrast, focus and consistent spacing.
2. Observed infrastructure	Build Overview and Site explorer using cleaned data. Confirm filters, selections, totals, missing values and source dates agree.
3. Map integration	Add OSM-based map, clustering and study bounds. Check attribution remains visible and table fallback works.
4. Research results	Add models only after target approval and evaluation. Verify units, held-out metrics, baseline comparison and chart labels.
5. Demonstration	Test one complete journey: filter > select site > inspect > review model > export. Test empty data and failed-map states.

Design handoff checklist

Create Figma frames at desktop 1440 x 900 and mobile 390 x 844. Include Overview, Site explorer, Model analysis and Data & methods; show default, selected-site, loading, empty and error states. Name reusable components and use the tokens from page 02. Verify real content fits before presenting screenshots.

Official references and implementation notes

These references support the implementation and accessibility notes. The palette, dimensions and screen concepts are original design recommendations.

[1] OpenStreetMap copyright and attribution

https://www.openstreetmap.org/copyright

[2] OpenStreetMap Foundation tile usage policy

https://operations.osmfoundation.org/policies/tiles/

[3] W3C - contrast minimum (text)

https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html

[4] W3C - non-text contrast

https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html

References checked 26 October 2026. Follow the chosen tile provider's own attribution and service conditions as well. The map drawings in this guide are schematic and contain no rendered OSM geography.